MW-A-001 INSPECTION PLAN

1. First-article control plan

Lot: first tool trial, tooling change, cavity repair, resin change and process requalification.

Sample: one complete dimensional report per cavity plus five assembled sets unless engineering increases the sample.

Methods: calibrated CMM/vision system for profiles and coordinates; pin/plug gauges for bores; micrometer for walls; optical comparator for draft/radii.

Acceptance: drawing-specific limits control; reference dimensions are recorded but do not independently reject the lot.

Record actual values, equipment ID, calibration due date, operator, cavity, lot and disposition.

MW-A-001 INSPECTION PLAN

2. Feature-id inspection inventory

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-- MW-M-001A --
MW-M-001A-D01 | DATUM SYSTEM | A = inside rear axial stop; B = XZ split plane; C = side button axis |
    MODEL_XYZ; origin per design input | basic
MW-M-001A-E01 | BREP ENVELOPE | 26.988 x 15.850 x 110.000 mm | min=(-13.494,-0.550,-0.000) | reference
MW-M-001A-M01 | MOLD PULL / PARTING | +Y from XZ parting plane / two longitudinal edges on XZ plane |
    part-local | vendor DFM
MW-M-001A-M02 | GENERAL DRAFT / RADII | 1.5 deg; Rext>=0.8; Rint>=0.5 | all unspecified molded
    faces/edges | tooling tune
MW-M-001A-M03 | SIDE ACTIONS | press-to-arm aperture core; four screw counterbore/clearance cores if
    not drilled after molding | feature-specific | shutoff draft 3.0 deg target
MW-M-001A-W01 | WALL | 2.00 NOM / 1.80 MIN | radial | tooling tune
MW-M-001A-O01 | BUTTON THRU APERTURE | DIA 8.20 | XYZ=(0,+Y,72.00) | +/-0.15; draft 1.5 deg
MW-M-001A-O02 | GUARD / HEAD POCKET | OD 12.80; pocket DIA 11.20 | XYZ=(0,+Y,72.00) | 0.40 diametral
    head clearance
MW-M-001A-H01 | 4X M2 CLEARANCE | DIA 2.30 | X=[-12.15, 12.15]; Z=[40.0, 90.0] | H11 unless supplier
    changes
MW-M-001A-H02 | 4X COUNTERBORE | DIA 4.20 x 1.50 deep | X=[-12.15, 12.15]; Z=[40.0, 90.0] | +0.10/-0
MW-M-001A-F01 | SEAM TONGUE / FIT | W 1.20 x D 0.55 | XZ parting edges | 0.10/side nominal
-- MW-M-001B --
MW-M-001B-D01 | DATUM SYSTEM | A = inside rear axial stop; B = XZ split plane; C = carrier rail |
    MODEL_XYZ; origin per design input | basic
MW-M-001B-E01 | BREP ENVELOPE | 26.988 x 13.500 x 110.000 mm | min=(-13.494,-13.500,-0.000) |
    reference
MW-M-001B-M01 | MOLD PULL / PARTING | -Y from XZ parting plane / two longitudinal edges on XZ plane |
    part-local | vendor DFM
MW-M-001B-M02 | GENERAL DRAFT / RADII | 1.5 deg; Rext>=0.8; Rint>=0.5 | all unspecified molded
    faces/edges | tooling tune
MW-M-001B-M03 | SIDE ACTIONS | USB-C service window core; debug window core; four tapered M2 thread-
    forming pilot cores | feature-specific | shutoff draft 3.0 deg target
MW-M-001B-W01 | WALL | 2.00 NOM / 1.80 MIN | radial | tooling tune
MW-M-001B-O01 | USB-C OPENING | 10.00 x 4.80 R1.00 | XYZ=(0,-Y,47.00) | profile +/-0.15
MW-M-001B-O02 | DEBUG OPENING | 8.00 x 3.20 R0.80 | XYZ=(0,-Y,60.00) | profile +/-0.15
MW-M-001B-H01 | 4X TAPERED PILOT | DIA 1.80->1.60 | X=[-12.15, 12.15]; Z=[40.0, 90.0] | supplier
    strip-torque validation
MW-M-001B-F01 | SEAM GROOVE / TONGUE | tongue W 1.20 | XZ parting edges | 0.10/side nominal
MW-M-001B-F02 | CARRIER KEY RAIL | tip Y -6.70; W 3.00 | X=0, Z=13..107 global | 0.20 nominal
-- MW-M-002 --
MW-M-002-D01 | DATUM SYSTEM | A = PCB support ledges; B = rear stop; C = key groove center |
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3. Assembly verification

Verify BOM identity and quantity closure.

Verify each released transform against assembly datum features.

Run positive-volume interference analysis; no unexpected interference is permitted.

Confirm controlled openings are unobstructed and all keep-out rules are satisfied.

Retain drawing, native-reopen, interference and inspection evidence under the same revision hash set.